ENGR 202 Week 7 Project Plan

Lab 6 Characterizing Frequency Dependent Circuits

Meeting dates: Thursday November 12 and Friday November 13, 2026
Time: 6:00-8:00 PM on both days, Pacific Standard Time (Oregon local time).
Group: To be confirmed Members: A / B / C; names to be confirmed.

Project objective

Characterize the RC low-pass and RL high-pass filters using analytical predictions, measured frequency responses, Excel Bode plots, and LTspice AC simulations. Complete all 19 questions with traceable evidence and explain any measured deviation from the ideal models.

Responsibilities

Owner	Questions and responsibilities	Required handoff
A RC lead	Q3.1-3.7 and Q3.9. Own RC setup, theory, hand sketches, measurements, Excel plots, and comparison.	RC data, setup photograph, cursor captures, derivations, hand plots, and comparison text.
B RL lead	Q4.1-4.7 and Q4.9. Own the equivalent RL workflow and investigate nonideal behavior.	RL evidence package; record inductor DC resistance to assess one possible nonideal effect.
C Simulation and integration	Q3.8 and Q4.8. Support Excel overlays, review units and phase signs, assemble the report, and coordinate Q1.1.	Both AC simulations, reviewed overlays, organized appendices, and a complete report draft.

All members confirm Q1.1 and review the final report. Member names and the group leader will be confirmed by the team.

Preparation before Thursday

A and B prepare their filter classification, transfer-function derivation, numerical expressions, and hand-drawn Bode plots. C prepares both ideal LTspice AC sweeps and shared data tables. Confirm component values, instrument availability, and file naming before the first meeting.

Meeting schedule

Meeting	Agenda and checkpoint
Thursday November 12 6:00-8:00 PM	6:00-6:15: Confirm roles, wiring, grounding, and measurement settings. 6:15-7:20: A operates the RC sweep; B records values and checks cursors; C checks simulation and chart consistency. 7:20-7:45: Review RC gain, phase, cutoff, slope, and screenshot completeness; repeat questionable points. 7:45-8:00: Save the RC evidence package and prepare the RL setup.
Friday November 13 6:00-8:00 PM	6:00-6:10: Verify RL setup and inductor resistance. 6:10-7:15: B operates the RL sweep; A records values and checks cursors; C updates simulations and charts. 7:15-7:45: Compare theory, simulation, and measurements for both filters; discuss discrepancies and resolve missing evidence. 7:45-8:00: Assemble and jointly review the report; record any remaining corrections.

Measurement and completion criteria

Use CH1 = Vin and CH2 = Vout, with a 2 Vpp sine input. Measure 10, 20, 50, 100, 200, 500, 1000, 2000, 5000, 10000, and 20000 Hz for each filter, adding points near cutoff. Every point requires input/output RMS and Vpp readings plus cursor-based phase evidence. Use Appendix B for RC and Appendix C for RL.

Before submission, verify both setup photographs, derivations, hand sketches, data tables, Excel plots, AC simulations, and comparisons. A and B approve their respective sections; C integrates corrections. The designated group leader submits by the actual Canvas deadline; this plan does not set a new deadline.